

Eclipse IDE - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://www.eclipse.org/>

<https://help.eclipse.org/>

<https://www.eclipse.org/downloads/>

Prerequisites

Use a JDK version supported by the project. Eclipse may bundle a runtime for the IDE, but your Java project should still use the JDK required by its stack.

Windows installation

Download Eclipse Installer from the official site, run it, choose Eclipse IDE for Java Developers or Enterprise Java and Web Developers, select the install folder and finish. Then choose a workspace.

QA setup

Configure the JDK under Installed JREs, import the Maven/Gradle project, resolve dependencies, configure Git and run a test class. Use breakpoints and Debug to investigate failures.

Validation and troubleshooting

Check project JDK, encoding, corporate proxy, Maven/Gradle and dependency resolution. If compilation fails, validate Java and the build outside the IDE first.

How to start safely

Start from the official Eclipse IDE site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Eclipse IDE, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Common issues and troubleshooting

When Eclipse IDE fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.